Algorithm 1 System Controller Logic

loop
  Read sensors: $T_{storage}$, T_{PVT}, $V_{storage}$, LDR (Sunlight)
  Read external forecaster (Weather)
  if $\min(T_{storage}, T_{PVT}) \leq T_{freeze}$ **then**
    {Case 4: Freeze Protection}
    Open storage valve; Close supply valve
    Turn heater ON
    continue
  end if
  if $\max(T_{storage}) > T_{max\_safe}$ **then**
    {Case 5: Overtemperature Emergency}
    Close all valves; Turn heater OFF
    continue
  end if
  if $V_{supply} < V_{refill\_threshold}$ **then**
    {Case 3: Emergency Supply Refill}
    Open supply valve; Open storage valve
    if Predicted mixed temp $< T_{minimum}$ **then**
      Turn heater ON
    end if
    continue
  end if
  if $T_{PVT} \geq T_{PVT\_ready}$ **then**
    if $V_{storage}$ is FULL **then**
      {Case 8: Storage Full}
      Close all valves
    else
      {Case 2: Fill Storage}
      Open supply valve; Open storage valve
    end if
  else if Supply Valve is CLOSED **and** Sun is OUT **and** Forecast is GOOD **then**
    {Case 1: Start PVT Flow}
    Open supply valve; Close storage valve
  else if Sun is NOT OUT **and** Cloud Duration > Cloud Tolerance **then**
    {Case 9: Prolonged Cloud Cover}
    Close all valves
  end if
  {Heater Control (Cases 6 & 10)}
  if $T_{avg} \leq T_{heater\_on}$ **or** (No Sun Expected **and** $T_{avg} < T_{target}$) **then**
    Turn heater ON
  else if $T_{avg} \geq T_{heater\_off}$ **then**
    Turn heater OFF
  end if
  {Case 7: Passive / Night Mode}
  if Sun is NOT OUT **and** Supply valve is CLOSED **then**
    Close storage valve
  end if
end loop